

Chapter 8, Problem 4RQ

Problem

If the roots of the characteristic equation of an *RLC* circuit are -2 and -3 , the response is:

(a) $(A \cos 2t + B \sin 2t)e^{-3t}$

(b) $(A + 2Bt)e^{-3t}$

(c) $Ae^{-2t} + Bte^{-3t}$

(d) $Ae^{-2t} + Be^{-3t}$

where A and B are constants.

Step-by-step solution

Step 1 of 3

The roots of the characteristic equation of an *RLC* circuit are,

$$s_1 = -2$$

$$s_2 = -3$$

The roots of the characteristic equation of an *RLC* circuit are real and different. So, the system is overdamped system.

Step 2 of 3

Option (a) is the general expression for the underdamped system, where the roots are complex. So, option (a) is wrong.

Option (b) is the general expression for the critical damped system, where the roots are real and equal. So, option (b) is wrong.

Option (c) is not general expression. So, option (c) is wrong.

Step 3 of 3

The general expression for the overdamped system is,

$$Ae^{s_1 t} + Be^{s_2 t} = Ae^{-2t} + Be^{-3t}$$

Thus, the correct answer is **d**.